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For fulfillment of requirements for the award of degree of pharmacy
(B. Pharm.)

Analytical Method Development and Validation of Mirabegron By Using Rp-HPLC Method of Bulk and Pharmaceutical Dosage Form.

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INTRODUCTION

Analytical chemistry is a science and art of recognized the various substance use in specify matrix and determination of its constitute.

Analytical chemistry is derived in two parts:

- Qualitative analysis
- Quantitative analysis

Chromatography:
A method of separating a mixture of components into individual components through equilibrium distribution between two phases that is mobile phase and stationary phase.
There are two types of chromatography:

- Adsorption Chromatography
- Partition Chromatography

High Liquid Performance Chromatography:

- HPLC is an advanced form of chromatography.
- It has mobile phase and stationary phase.
- A sample mixture in a solvent(mobile phase) is pumped at high pressure through a column with packing material(stationary phase).

Advantages:

- High accuracy
- High reproducibility
- Fast analysis

Applications:

- Widely used for drug analysis and drug separation.
- Separates compounds based on polarity and interactions.

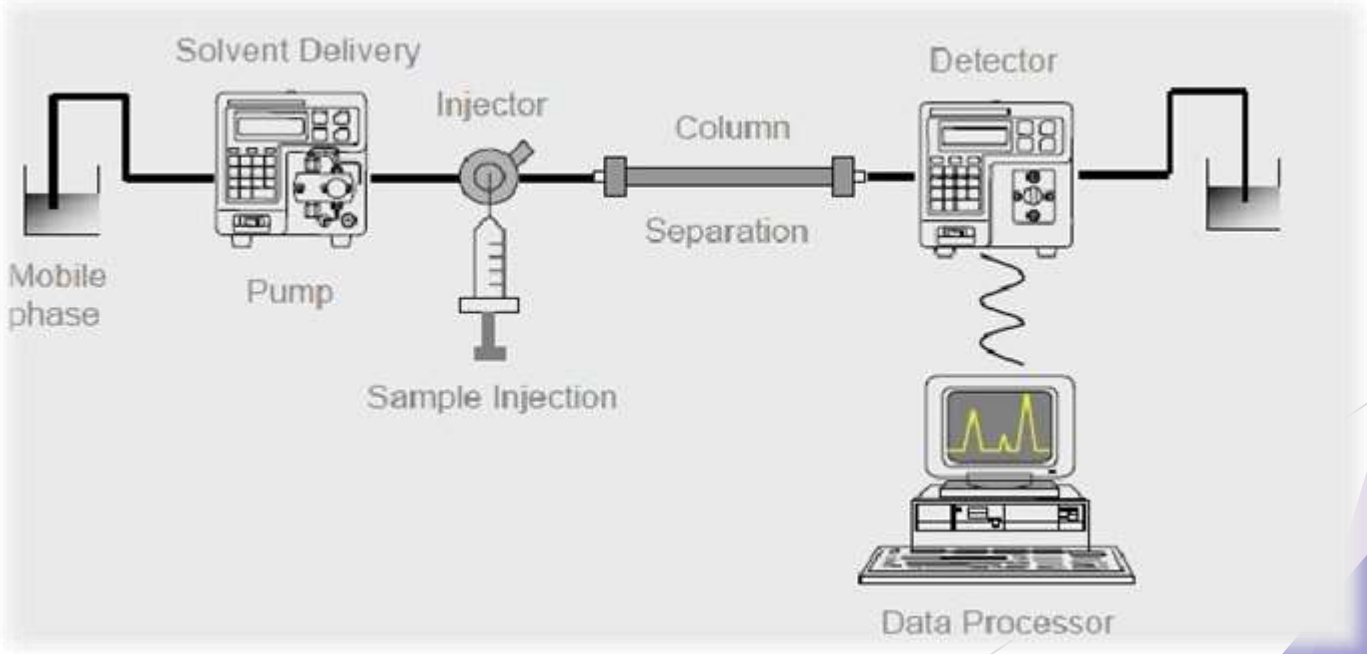
Instrumentation:

1. Solvent reservoir and Treatment system

✓ Following are the components in HPLC:

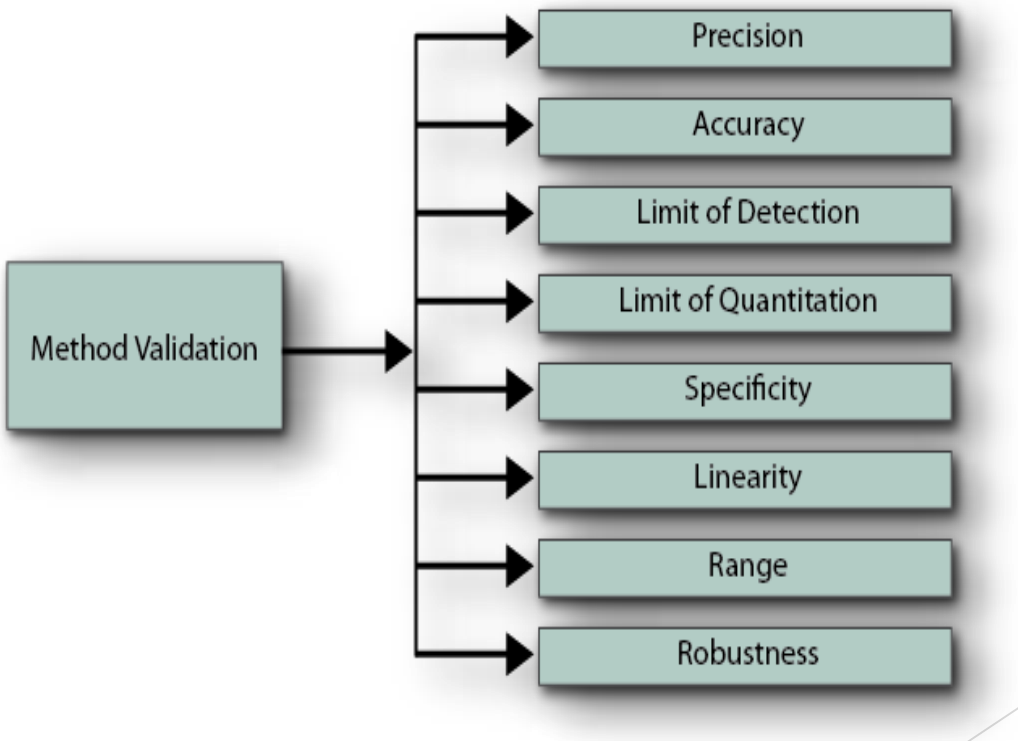
1. Mobile phase
2. Pump system
3. Sample Injection System
4. Column
5. Detector

Diagram of HPLC



Validation:

- Assures that the method works reproducibly.
- To understand parameters in the process.
- ICH Guidelines.



Methods of validation

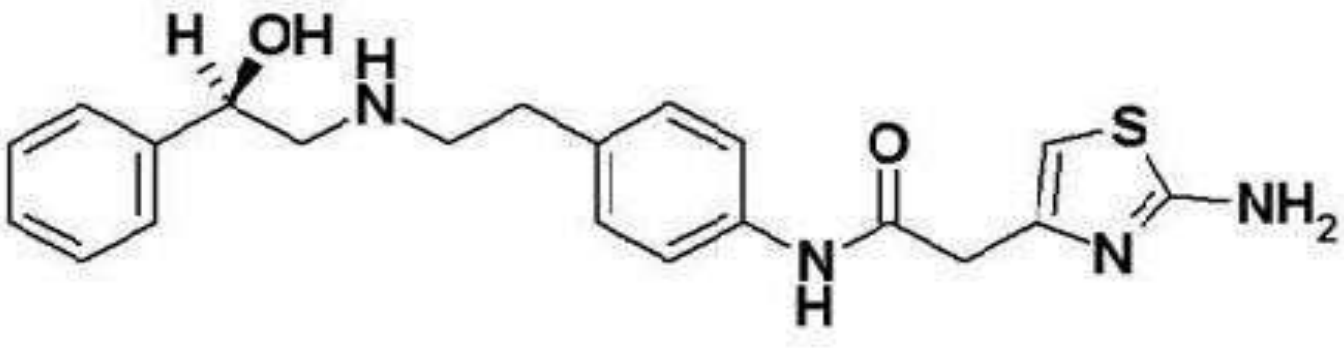
OBJECTIVES

The objectives of this dissertation work areas follows:

- ✓ To develop simple, rapid and sensitive, stable and highly effective simple RP-HPLC method for determination of Mirabegron.
- ✓ To validate methods as per ICH Guidelines.
- ✓ To analyse Mirabegron using validated methods quantitatively and qualitatively.

DRUG PROFILE

- Drug: Mirabegron
- Brand name: Miraben ER-25
- Structure:



Structure of mirabegron

- Molecular Formula: $C_{18}H_{22}N_2O_2$
- Molecular Wt.: 298.4 g/mol
- Category: Beta-3 adrenergic receptor
- Use: To treat Overactive bladder (OAB)

EXPERIMENTAL WORK

HPLC was selected as analytical technique for estimation of Mirabegron.
Trail and error method was used for the selection of chromatographic conditions.

Sr no.	Parameters	Description
1.	HPLC	Agilent Tech. Gradient System With Auto injector, UV Detector
2.	Software	Chemstation 10.1
3.	Column	(Agilent) C18 column (4.6 x 150mm)
4.	Particle size	5µm
5.	Stationary phase	C-18 (Agilent)
6.	Mobile phase	Methanol:0.1% OPA (55:45)
7.	Detection wavelength	254nm
8.	Flow rate	0.7 ml/min
9.	Temperature	Ambient
10.	Sample size	20µl
11.	Ph	3

Sr no.	Parameters	Description
12.	Run time	15 min
13.	Filter paper	0.45µ

- UV-VIS Spectrophotometer:
  UV-VIS Spectrophotometer was selected for estimation.
  UV Visible absorbance range of 200-400nm.
  The instrument needs about 1 minute for initialization.

- Light Source used:
  Deuterium lamp

- Cuvette used:
  Quartz cuvette

- Preparation of Stock Standard Solution:

Standard Solution Stock I : (Mirabegron)

Accurately weigh and transfer 5 mg Mirabegron working standard into 10 ml volumetric flask as about diluents Methanol completely and make volume up to the mark with the same solvent to get 500 µg/ml standard (stock solution) and 15 min sonicate to dissolve it and the resulting stock solution 0.1ml was transferred to 10 ml volumetric flask and the volume was made up to the mark with mobile phase Methanol: (0.1%OPA)Water, prepared in (5.5 ml MEOH: 4.5 ml (0.1%OPA)Water v/v)solvent .

- Validation method:

ICH Guidelines were used for the method validation guidelines Q2 (R1).

RESULT AND DISCUSSION

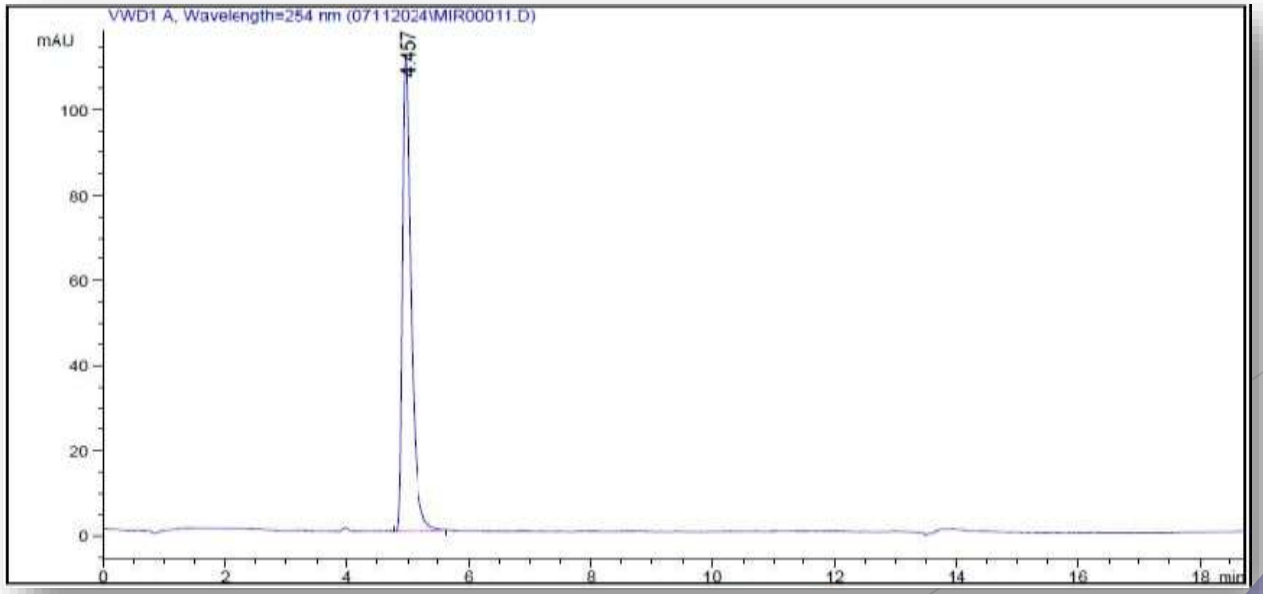
Preliminary studies on Mirabegron were done.

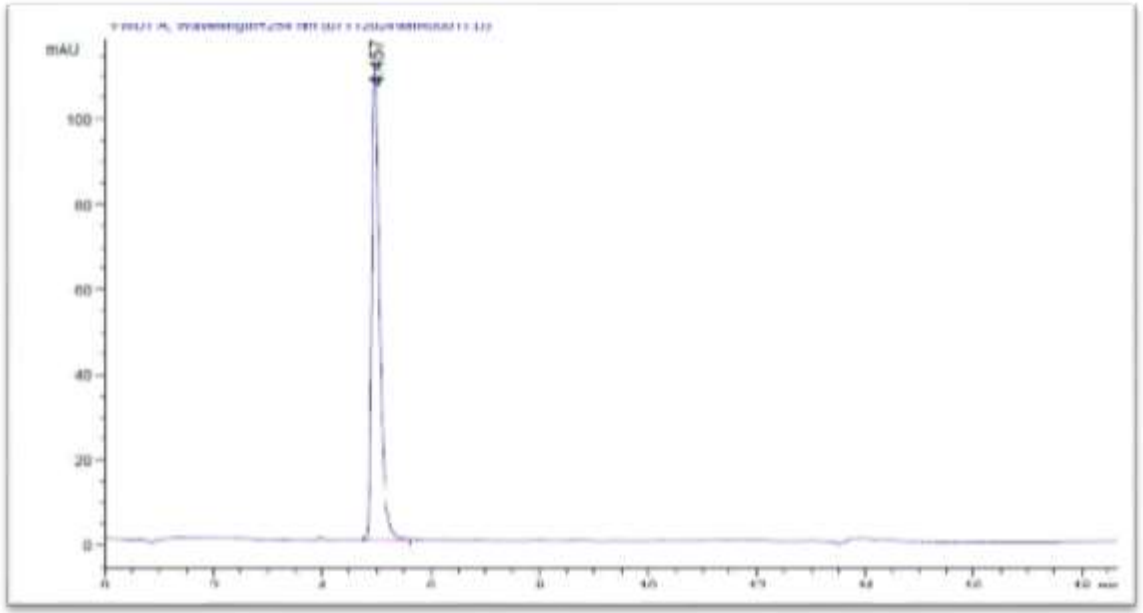
- Melting point
- Appearance
- Solubilty

➢ Studies on the chromatographic behavior of Mirabegron:

After the selection of suitable mobile phase, it was then optimized for its reproducibility, sensitivity & accuracy.

Wavelength curve of mirabegron





RT	Area	TP	TF	Resolution
4.469	368.5	7846	0.62	-

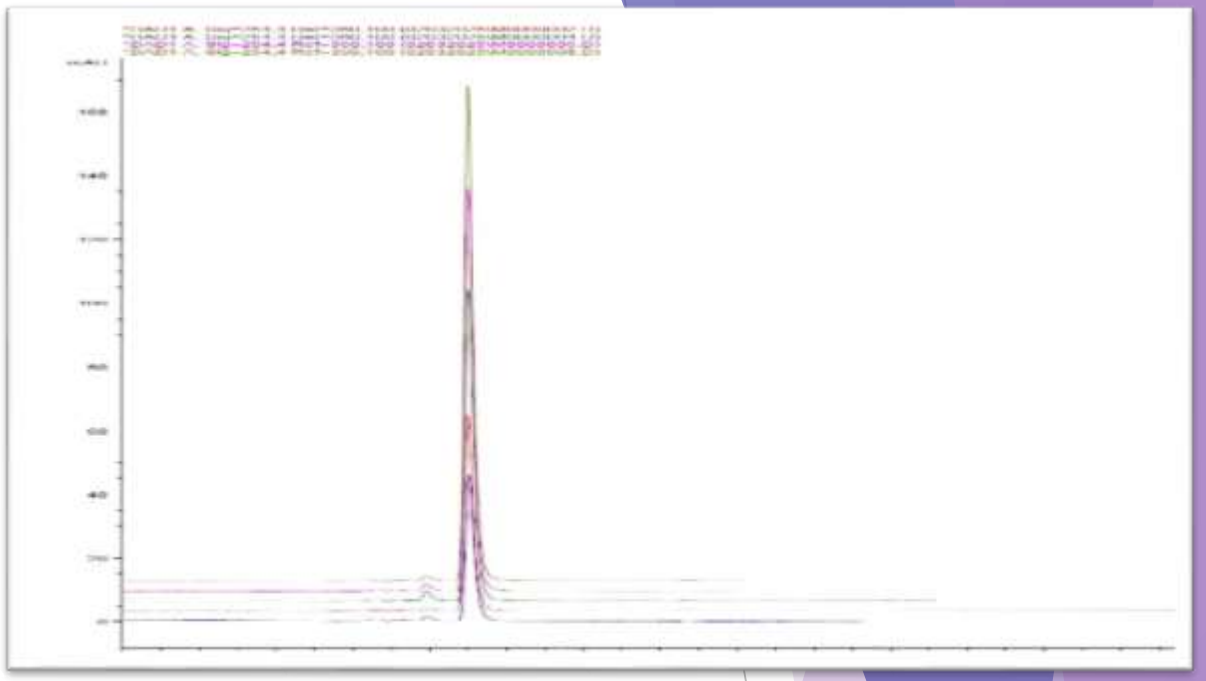
Chromatography curve of Final trail

Analytical of Method Validation:

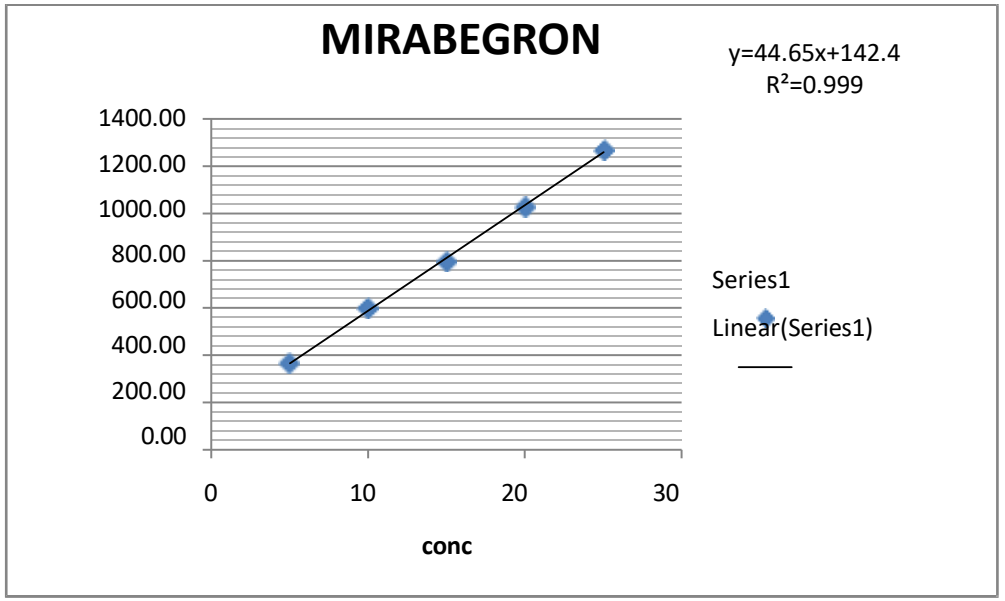
1. **Linearity:**
   From Mirabegron standard stock solution different working standard solution (5-25 µg/ml) were prepared in mobile phase 20 µl of sample solution was injected into the chromatographic system using mixed volume loop injector. Chromatograms were recorded.

Sr no.	Conc	Area MBG
1.	5	368.09
2.	10	597.48
3.	15	789.30
4.	20	1028.49
5.	25	1268.83

Result of linearity of chromatography



Overlay of linearity(5-25mcg)



Calibration curve of MBG

Regression equation data Y=mx+c	
Slope(m)	44.65
Intercept(c)	142.4
Correlation Coefficient	0.999

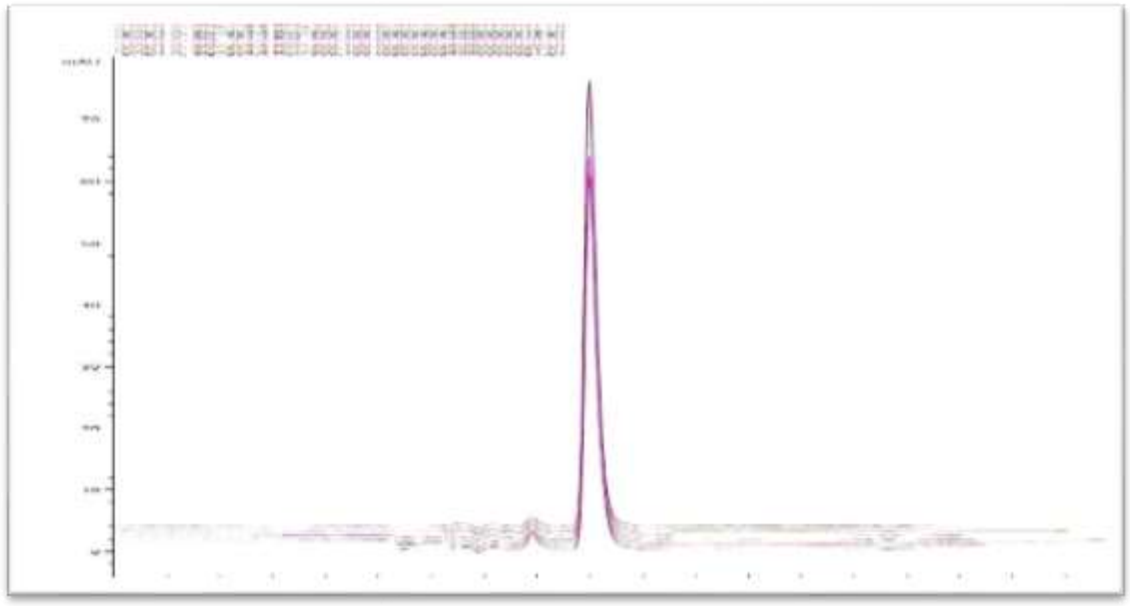
2. Accuracy:

To pre-analyzed Tablet solution, a definite concentration of standard drug (80%, 100%, and 120%) was added and then its recovery was analyzed

Sr no.	Level (%)	Amt. taken	Amt. added	Area mean *±S.D	Amt recovered mean*±S.D	% Recovery mean* ± S.D
1.	80%	5	4	8.98±0.05	3.9±0.051	99.59±1.28
2.	100%	5	5	9.97±0.030	4.97±0.030	99.45 ± 0.60
3.	120%	5	6	11.0±0.078	6.03±0.078	100.5 ±1.31

Result of recovery data for MBG

Overlay of Accuracy(5-25mcg)



Accuracy of method by recovery studies was found to be between 98-102%

3. Repeatability:

To ascertain the resolution and reproducibility of the proposed chromatographic system for estimation of Mirabegron system suitability parameters were studied.

Sr no	Conc. of MBG(mg/ml)	Peak area	Amt. found(mg)	% Amt. found
1.	15	800.87	800.77	98.30
2.	15	800.66		
3.	15	801.868		
		Mean	14.75	100.32
		SD	0.15	0.15
		%RSD	0.02	0.02

Repeatability studies of Mirabegron was found to be 98.30 ,The % RSD was less than 2, which shows high percentage amount found in between 98% to 102% indicates the analytical method that concluded .

4. Precision:

All the solution was analyzed thrice in order to record any intra-day & inter-day variation in the result that concluded.

Conc. (µg/ml)	Intra-day precision			Inter-day precision		
	Mean±SD	%Amt Found	%RSD	Mean±SD	Mean±SD	%RSD
10	597.17± 0.03	101.85	0.00	595.88 ±1.08	101.56	0.18
15	798.73± 0.06	98.00	0.06	804.94 ±1.44	98.92	0.18
20	1029.33 ±0.18	99.32	0.18	1032.44± 1.44	99.67	0.18

Intraday and Interday Precision for Mirabegron which shows the high precision % amount in between 98% to 102% indicates to analytical method that concluded.

5. Robustness:

To evaluate it small but deliberate variations in the optimizsed methods parameters where done. The effect of changes in mobile phase composition and flow rate, wavelength on retention time and tailing factor of drug peak was seen.

Parameters	Conc. (µg/ml)	Amt. detected	%RSD
Mobile phase 54ml+46ml	15	801.68 ± 0.98	0.12
Mobile phase 55ml+45ml	15	803.7±2.02	0.25
253 nm	15	747.5±1.44	0.19
254 nm	15	864.89±1.70	0.20
Flow rate 0.6ml	15	940.23±1.20	0.13
Flow rate 0.8ml	15	702.79±3.46	0.49

The changes were done flow rate (±1 ml/ min^{-1}), PH of mobile phase composition, and Wavelength. %RSD for peak area was calculated which should be less than 2%.the result shown in analytical method that concluded.

6. Limit of Detection:

Lowest limit that can be detected.

LOD of MBG was found to be 0.0572(µg/ml)

7. Limit of Quantification:

Lowest concentration that can be quantitatively measured.

LOD of MBG was found to be 0.1735 (µg/mL)

- **Analysis of Tablet formulation:**

Weigh 20 Mirabegron Tablet and calculate the average weight, accurately weigh and transfer the sample equivalent to 16.89 mg Mirabegron into 10 ml volumetric flask. Add about 10 ml of diluents and sonicate to dissolve it completely and make volume up to the mark with diluent. Mix well and filter through 0.45 µm filter. Further pipette 0.4 ml of the above stock solution into a 10 ml volumetric flask and dilute up to the mark with diluents. (20 µg/ml). The simple chromatogram of test Mirabegron was done and the amounts of Mirabegron per Tablet were calculated by extrapolating the value of area from the calibration curve.

- **Brand Name Mirabegron ER-25(Prevego Health Care)**

Total weight of 20 tablets= 16.89 gms
Avg. powder wt.= 0.8445 gm/ tab
Eq. wt. for 5 mg= 16.89 mg
Take 16.89 mgs in 10 ml methanol = 500 µgm/ml

Sample	Lable claim	% Lable claimed ± SD	% RSD
Mirabegron	25 mg	99.41±0.071	0.072

%Lable claim

Tablet Assay for % Lable claim for was also was found to be 99.80%and %RSD are less than 2 satisfactory result was concluded.

9. Ruggedness:

Analyst	RT	Area	Therotical plates	Symmetry
1.	4.455	593.101	7818	0.69
2.	4.460	594.11	7649	0.62

Analysis of Ruggedness

SUMMARY

- Mirabegron is phosphodiesterase-5 inhibitor (PDE5) drug, the present work deals with “Analytical method development and validation of Mirabegron by Using RP-HPLC method of bulk and pharmaceutical dosage form.”

- Attempts were made to develop RP-HPLC method for simultaneous estimation of Mirabegron from Tablet. For the RP -Agilent Tech. Gradient System with Auto injector, UV (DAD) & Gradient Detector Reverse Phase (Agilent) C18 column (4.6mm x 150mm;5µm), a 20µl injection loop and UV730D Absorbance detector and running chemstation 10.1 software.

- Methanol: water (0.1%OPA), (55:45) v/v, pH 3.was used as the mobile phase for the method. The detection wavelength was 254 nm and flow rate was 0.7 ml/min. In the developed method, the retention time of Mirabegron were found to be 4.457 min. The developed method was validated according to the ICH guidelines. The linearity, precision, range, robustness were within the limits as specified by the ICH guidelines. Hence the method was found to be simple, accurate, precise, economic and reproducible.

CONCLUSION

- This developed RP-HPLC method for estimation of Mirabegron is accurate, precise, robust and specific.

- The method has been found to be better than previously reported method, because of its less retention time, isocratic mode and use of an economical and readily available mobile phase, readily available column, UV detection and better resolution of peaks.

- The PDE5 agents of Mirabegron results from a decrease in systemic vascular resistance and the parent compound Mirabegron is primary responsible for the is phosphodiesterase- 5 inhibitor.

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THANK YOU